

G-01-L3-concentration v1: L3 -j local critical concentration

ZFL Lab

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INPUT: $C_{L^3} = \text{PROVEN}$: $T_* < \infty \implies \limsup_{t \rightarrow T_*} \|u(t)\|_3 = +\infty$. C_E^{weak}, C_E^{strong} OPEN/BLOCKED as before. $\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}\}$.

FORBIDDEN: Assume CKN ε -regularity without proof. Assume spatial concentration by heuristic scaling. Assume C_G to prove C_G .

MISSION: Audit whether C_{L^3} forces existence of critical local concentration.

Candidate C_G :

There exists $\epsilon_0 > 0$ (universal) and sequences $t_n \uparrow T_*$, $x_n \in \mathbb{R}^3$, $r_n \downarrow 0$ such that for $B_n := B(x_n, r_n)$,

$$\|u(t_n)\|_{L^3(B_n)} \geq \epsilon_0$$

or scale-invariant local quantity

$$r_n^{-2} \int_{B_n} |u(t_n)|^3 \geq \epsilon_0^3 \text{ or } r_n^{-1} \|u\|_{L^3(B_n)}^3 \geq \epsilon_0$$

with precise scaling to be fixed, and $r_n \sim \sqrt{T_* - t_n}$ if parabolic.

Why this is non-trivial:

$\|u(t)\|_3 \rightarrow \infty$ in limsup is global. Does not a priori imply mass piles at small scale; could spread. To get concentration, need profile decomposition / compactness up to scaling for Navier-Stokes.

Known results: ESS / Kenig-Koch non-concentration vs concentration: For Leray-Hopf, if $T_* < \infty$, then there exists such concentration (Escauriaza-Seregin-Sverak type). But proof uses backward uniqueness, Carleman, ε -regularity. We cannot assume; we must audit steps.

Steps to PROVE C_G from C_{L^3} :

1. Rescaling: Suppose $\|u(t_n)\|_3 = M_n \rightarrow \infty$. Define $v_n(y) := r_n u(x_n + r_n y, t_n)$ with $r_n = M_n^{-?}$ to normalize L^3 . Scale invariance: $\|v_n\|_3 = \|u(t_n)\|_3$ global, but local norm $\|v_n\|_{L^3(B_1)}$ should carry lower bound.

2. Need non-vanishing lemma: $\limsup \|u(t_n)\|_3 = \infty$ implies after suitable translation+scaling, weak limit non-zero. This is profile decomposition for L^3 (critical). Requires.

3. Then ϵ_0 appears as minimal L^3 mass of non-trivial profile.

Thus C_G is not immediate from C_{L^3} by Holder; needs compactness.

Output of this audit:

$C_{L^3} \implies C_G$ NOT PROVEN from current hypotheses alone

C_G = concentration local critical candidate OPEN, but precisely formulated

C_G requires: profile decomposition / ε -regularity / ESS-concentration theorem

Consequence:

Current global $\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}\}$ does NOT yet yield C_G . The bridge is identified:

$C_{L^3} + (\text{backward uniqueness} / \text{Carleman} / \text{local smoothing}) =_i C_G$ PROVEN.

Therefore next experiment should be:

G-02: formalize ε_0 threshold from literature (ESS 2003) as external theorem to import, with clear dependency tag, not as internal proof, to keep ZFL traceability.

No $\perp$ from $C_E^{weak} \wedge C_{L^3}$. Need C_G .

Status: $C_G = \text{OPEN}$ with precise candidate definition.